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## *Research Statement*

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My research asks how to make modern machine learning systems more data-efficient, computationally efficient, and reliable. As models grow in scale and are deployed across diverse tasks, the cost of training and adaptation increasingly depends not only on model architecture, but also on which data are used, how optimization is performed, and how knowledge is shared across tasks. My work develops principled methods for these problems, combining ideas from optimization, online learning, submodular analysis, optimal transport, and statistical modeling with empirical evaluation on contemporary machine learning systems.

A central premise of my research is that more data and computation are not always necessary to achieve better learning. Instead, carefully choosing informative examples, allocating limited training budgets, and exploiting structure among tasks can make learning substantially more efficient. My research has explored these ideas in data subset selection for language models and vision models, parameter-efficient adaptation and knowledge transfer, prototype selection, and personalized federated learning. Across these settings, I aim to connect algorithmic insight with practical methods that can be evaluated at realistic scales.

## Data Selection and Optimization for Efficient Learning

Large-scale machine learning often treats the training dataset as fixed and processes examples using generic sampling strategies. Yet examples differ in their value to a learning objective, and the best subset may depend on the model, the optimization procedure, the target task, and the available compute budget. This motivates a central question in my work: *How can we select and organize training data so that a limited amount of computation produces the greatest learning progress?*

One direction of my research develops principled methods for adaptive subset selection. In my work on *Bandit-Guided Submodular Curriculum for Adaptive Subset Selection* (NeurIPS 2025), I study how to combine combinatorial objectives with bandit feedback to select informative subsets over the course of training. The framework considers objectives such as facility location, graph cut, log-determinant diversity, and mutual information, while using validation performance to guide adaptive decisions. The broader goal is to move beyond a one-time selection of a static coreset and instead adapt selection as the model and its learning needs evolve.

A second direction concerns selecting minibatches for language-model training. In *Minibatch Selection for Language Models via Partition Matroid Constrained Gradient Matching* (ICML 2026), I study how to choose training examples whose gradients best represent a target update while respecting partition-based selection budgets. The method connects gradient matching to constrained combinatorial optimization, using structure in the selection problem to develop an efficient algorithm. This line of work reflects my broader interest in making data selection compatible with the practical constraints of modern training pipelines, including limited selection budgets and the need to keep selection overhead manageable.

These projects raise fundamental questions about the relationship between data utility and optimization. A subset that approximates gradients under one optimizer may not be equally useful under another, and selection criteria based on local gradient information may fail to capture longer-term training effects. My ongoing work explores how the geometry and dynamics of optimization should influence data selection, including settings where optimizers transform gradients before applying parameter updates. This perspective opens a route to selection methods that are tailored to the learning algorithm rather than designed independently of it.

**Future directions.** I plan to investigate data selection objectives that account for optimizer dynamics, training stage, and downstream generalization. Important questions include how to characterize the value

of a subset under non-Euclidean or structured updates; how to balance representativeness, diversity, and learning progress; and how to obtain guarantees when the selection objective is only weakly submodular, noisy, or accessible through expensive feedback. I am also interested in selection methods that work across modalities and training regimes, and in understanding when adaptive selection offers meaningful advantages over static coresets. My goal is to develop algorithms with clear theoretical motivation and measurable end-to-end gains, including accounting for the computational cost of selection itself.

## Efficient Adaptation and Knowledge Transfer in Large Models

Large pretrained models provide powerful general-purpose representations, but adapting them to new tasks can still require substantial data, compute, and memory. In addition, training examples vary in quality and relevance, and the cost of adapting a model can become prohibitive when many tasks or domains must be supported. My research in efficient learning therefore also examines how to reduce the cost of model training and adaptation through parameter-efficient methods, knowledge transfer, and better use of training data.

During my research work at Google DeepMind, I worked on efficient language-model pretraining and knowledge distillation, including methods for routing and transferring information between models. This experience shaped my interest in treating efficiency as an algorithmic problem rather than only a systems optimization problem: which computations are necessary, which examples are useful, and what information should be retained when a model is compressed or adapted?

My research on data selection for language models complements this direction. By choosing informative examples and matching updates to a target learning signal, it is possible to reduce the amount of training required for a given objective. More broadly, I am interested in how data selection, parameter-efficient fine-tuning, and knowledge distillation can be designed jointly. For example, a subset that is useful for full-parameter training may not be optimal for low-rank adaptation, and the best distillation examples may depend on which capabilities a student model needs to preserve.

**Future directions.** I aim to develop principled methods for efficient pretraining and adaptation that jointly consider data, parameters, and computation. This includes selecting data for parameter-efficient fine-tuning, identifying examples that best transfer knowledge from teacher to student models, and studying the interaction between model architecture, optimizer choice, and selection strategy. I am particularly interested in methods that retain strong performance without relying on expensive model-scale experimentation at every iteration, and in theoretical frameworks that explain when particular forms of parameter sharing or data reduction are effective.

## Structure-Aware Learning Across Tasks and Domains

A third theme in my research is how to exploit relationships among tasks, domains, or data sources to improve learning under limited budgets. In many practical settings, tasks are neither completely independent nor identical. Some share useful structure, while others require distinct information. Treating all tasks uniformly can waste data and computation; treating each task independently can discard valuable opportunities for transfer.

My work on *Learning Task Mixtures from Task Affinities: A Probabilistic Graphical Model for Supervised Fine-Tuning* studies how relationships among tasks can guide the allocation of training data for supervised fine-tuning. The central idea is to model task-level affinities explicitly and use them to inform task-mixture construction. This connects data allocation with structured probabilistic modeling: rather than choosing a training mixture using only task sizes or isolated utility estimates, we can reason about how tasks relate to one another and how those relationships affect a shared model.

I have also worked on prototype and subset selection using ideas from submodularity and partial optimal transport. In *UniPROT*, this perspective supports selecting representative prototypes while accounting for relationships between data points and the target structure. My earlier work on Bayesian coreset optimization for personalized federated learning similarly reflects a longstanding interest in selecting compact, informative

representations of data when communication, storage, or access to centralized data is constrained.

Together, these projects suggest a common research problem: *How can we allocate limited data and computation across related tasks while preserving the information needed for generalization?* The answer may require combining task-level structure with example-level utility, and connecting local selection decisions to global learning objectives.

**Future directions.** I plan to explore task-aware data allocation, structured mixtures for supervised fine-tuning, and prototype selection under distribution shift and resource constraints. One direction is to jointly optimize which tasks to include and which examples to select within each task. Another is to use task-affinity graphs to identify complementary or redundant data sources and to adapt the training mixture as evidence about downstream performance accumulates. I am also interested in fairness-constrained selection, where the chosen subset must satisfy demographic or group-level requirements in addition to utility and budget constraints. These problems create opportunities to combine combinatorial optimization, probabilistic modeling, optimal transport, and statistical learning theory.

## Long-Term Research Vision

My long-term goal is to build a principled foundation for data-efficient machine learning: algorithms that identify what information is needed for a learning objective, decide how limited computation should be allocated, and exploit structure across examples, tasks, and models. I see three closely related challenges.

First, I want to strengthen the theory of data selection. Existing methods often rely on assumptions about submodularity, gradient similarity, or access to a useful validation signal. Understanding when these assumptions are justified, how robust the resulting algorithms are, and how selection overhead affects total training cost is essential for translating theoretical improvements into practical benefits.

Second, I want to connect data selection more tightly with optimization and model adaptation. Selection, optimizer choice, and parameter-efficient fine-tuning are often studied separately, even though each changes the value of the others. A unified view could lead to methods that select examples for the updates a model will actually make, rather than for a proxy objective that may be poorly aligned with training dynamics.

Third, I want to develop methods for learning across related tasks under explicit resource and reliability constraints. Task affinities, prototype structure, and distributional relationships can guide how data and computation are shared, while constraints such as fairness, privacy, and limited communication determine which solutions are feasible. I am interested in algorithms that make these trade-offs explicit and offer useful guarantees without sacrificing empirical performance.

Methodologically, I aim to combine optimization and statistical analysis with carefully designed experiments on modern language models and other machine learning systems. I value research in which the problem formulation, algorithmic contribution, theoretical understanding, and empirical evaluation reinforce one another. Ultimately, I hope to help make high-performing machine learning more accessible by reducing the data and computational resources required to train, adapt, and deploy models reliably.